

Two & Four marks questions

1.	Explain the formation of linear programming problem.
2.	Define solution of the LPP.
3.	Define feasible solution of the LPP.
4.	Define basic feasible solution of the LPP.
5.	Define an optimal solution of the LPP.
6.	Define slack variable of the LPP.
7.	Define surplus variable of the LPP.
8.	Define the basic variables and non-basic variables in LPP.
9.	Define basic solution of the LPP.
10.	Explain the canonical form of LPP with an example.
11.	The XYZ company during the festival season combines two factors A and B to form a gift pack which must weigh 5 kg. At least 2 kg of A and not more than 4 kg of B should be used. The net profit contribution to the company is Rs. 5 per kg for A and Rs. 6 per kg for B. Formulate Linear programming model to find the optimal factor mix.
12.	A manufacturer produces two types of models M1 and M2. Each model of the type M1 requires 4 hours of grinding and 2 hours of polishing; whereas each model of the type M2 requires 2 hours of grinding and 5 hours of polishing. The manufacturer has 2 grinders and 3 polishers. Each grinder works 40 hours a week and each polisher works for 60 hours a week. Profit on M1 model is 3.00 and on model M2 is 4.00. Whatever is produced in a week is sold in the market. How should the manufacturer allocate his production capacity to the two types of models so that he may make the maximum profit in a week?
13.	A manufacturing company makes two types of television sets; one is black and white and the other is colour. The company has resources to make at most 300 sets a week. It takes Rs 1800 to make a black and white set and Rs 2700 to make a colored set. The company can spend no more than Rs 648000 a week to make television sets. If it makes a profit of Rs 510 per black and white set and Rs 675 per colored set, how many sets of each type should be produced so that the company has a maximum profit? Formulate this problem as a LPP given that the objective is to maximize the profit.
13.	A factory makes tennis rackets and cricket bats. A tennis racket takes 1.5 hours of machine time and 3 hours of craftsman's time in its making while a cricket bat takes 3 hours of machine time and 1 hour of craftsman's time. In a day, the factory has the availability of not more than 42 hours of machine time and 24 hours of craftsman's time. (i) What number of rackets and bats must be made if the factory is to work at full capacity? (ii) If the profit on a racket and on a bat is Rs 20 and Rs 10 respectively, find the maximum profit of the factory when it works at full capacity.

14.	A man rides his motorcycle at the speed of 50 km/hour. He has to spend Rs 2 per km on petrol. If he rides it at a faster speed of 80 km/hour, the petrol cost increases to Rs 3 per km. He has at most Rs 120 to spend on petrol and one hour's time. He wishes to find the maximum distance that he can travel. Express this problem as a linear programming problem.	
15.	Maximize $z = 3x_1 + 5x_2 + 7x_3$, subject to the constraints $6x_1 - 4x_2 \leq 5$, $3x_1 + 2x_2 + 5x_3 \geq 11$, $4x_1 + 3x_3 \leq 2$, $x_1 \geq 0$, $x_2 \geq 0$, convert the LPP to the standard form of the LPP.	
16.	Rewrite the following LPP's in Standard form:	
	$\begin{aligned} \text{Min } Z &= 2x_1 + x_2 + 4x_3 \\ \text{s.t. } -2x_1 + 4x_2 &\leq 4 \\ x_1 + 2x_2 + x_3 &\geq 5 \\ 2x_1 + 3x_3 &\leq 2 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$	$\begin{aligned} \text{Max } Z &= x_1 - 3x_2 \\ \text{s.t. } -x_1 + 2x_2 &\leq 15 \\ x_1 + 3x_2 &= 10 \\ x_1, x_2 &\text{ are unrestricted in sign} \end{aligned}$
	$\begin{aligned} \text{Min } Z &= -3x_1 + x_2 + x_3 \\ \text{s.t. } x_1 - 2x_2 + x_3 &\leq 11 \\ -4x_1 + x_2 + 2x_3 &\geq 3 \\ 2x_1 - x_3 &= -1 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$	$\begin{aligned} \text{Min } Z &= 2x_1 + x_2 + 4x_3 \\ \text{s.t. } -2x_1 + 4x_2 &\leq 4 \\ x_1 + 2x_2 + x_3 &\geq 5 \\ 2x_1 + 3x_3 &\leq 2 \\ x_1, x_2 \geq 0, x_3 &\text{ unrestricted in sign} \end{aligned}$
17.	Explain the working procedure of the graphical method to solve given LPP.	
18.	Solve the following linear programming problem graphically, minimize $z = 200x + 500y$ subject to the constraints $x + 2y \geq 10$, $3x + 4y \leq 24$, $x \geq 0$, $y \geq 0$.	
19.	Find the maximum value of $z = 3x_1 + 4x_2$ subject to the constraints $4x_1 + 2x_2 \leq 80$, $4x_1 + 2x_2 \leq 80$, $2x_1 + 5x_2 \leq 180$, $x_1, x_2 \geq 0$, by solving LPP using graphical method.	
20.	Find the maximum value of $z = 2x + 3y$ subject to the constraints $x + y \leq 30$, $y \geq 3$, $x - y \geq 0$, $0 \leq x \leq 20$ and $0 \leq y \leq 12$ using graphical method.	
21.	Minimize $z = 5x + 4y$ subject to the constraints $x + 2y \leq 10$, $x + y \geq 8$, $2x + y \geq 12$, $x \geq 0$, $y \geq 0$ by graphical method.	
22.	Explain working procedure of simplex method to solve the given LPP.	
23.	Define Artificial variables of the LPP.	
24.	Explain working procedure of Big M method or Method of penalties to solve the given LPP.	
25.	Explain working procedure of Two phase method to solve the given LPP.	
26.	Explain the concept of primal and dual of LPP.	
27.	State Fundamental theorem of Duality.	

28. Write the dual of the LPP to minimize $z = 2x_1 + 5x_2 + 6x_3$, subject to the constraints $5x_1 + 6x_2 - x_3 \leq 3$, $-2x_1 + x_2 + 4x_3 \leq 4$, $x_1 - 5x_2 + 3x_3 \leq 1$, $-3x_1 - 3x_2 + 7x_3 \leq 6$, $x_1 \geq 0$, $x_2 \geq 0$ and $x_3 \geq 0$.
29. Write the dual of the LPP to minimize: $z = 3x_1 - 2x_2 + 4x_3$ subject to $3x_1 + 5x_2 + 4x_3 \geq 7$, $6x_1 + x_2 + 3x_3 \geq 4$, $7x_1 - 2x_2 - x_3 \leq 10$, $x_1 - 2x_2 + 5x_3 \geq 3$, $4x_1 + 7x_2 - 2x_3 \geq 2$, $x_1, x_2, x_3 \geq 0$.
30. Write the dual of the LPP to maximize $z = 2x_1 + 3x_2 + x_3$ subject to $4x_1 + 3x_2 + x_3 = 6$, $x_1 + 2x_2 + 5x_3 = 4$, $x_1, x_2, x_3 \geq 0$.
31. Convert the LLP to dual form minimize: $z = x_1 + x_2 + x_3$ subject to $x_1 - 3x_2 + 4x_3 = 5$, $x_1 - 2x_2 \leq 3$, $2x_2 - x_3 \geq 4$, Where x_1, x_2, x_3 are unrestricted.

Seven to ten marks questions

32. Solve the following LLP by Graphical method:

$Max Z = 6x_1 + x_2$ $s/t \ 2x_1 + x_2 \geq 3$ $x_1 - x_2 \geq 0$ $x_1, x_2 \geq 0$	$Max Z = x_1 + 2x_2$ $s/t \ x_1 + x_2 \geq 3$ $x_1 - x_2 \leq 1$ $x_1, x_2 \geq 0$	$Max Z = x_1 + x_2$ $s/t \ x_1 + x_2 \leq 1$ $-3x_1 + x_2 \geq 3$ $x_1, x_2 \geq 0$
$Max Z = 2x_1 + 3x_2$ $s/t \ x_1 + x_2 \leq 30$ $x_1 - x_2 \geq 0$ $x_2 \geq 3$ $x_2 \leq 12$ $x_1 \leq 20$ $x_1, x_2 \geq 0$	$Max Z = 5x_1 + 3x_2$ $s/t \ x_1 + x_2 \leq 6$ $2x_1 + 3x_2 \geq 3$ $0 \leq x_1 \leq 3$ $0 \leq x_2 \leq 3$ $x_1, x_2 \geq 0$	$Max Z = 2x_1 + 4x_2$ $s/t \ x_1 + 2x_2 \leq 5$ $x_1 + x_2 \leq 4$ $x_1, x_2 \geq 0$
$Max Z = 6x_1 + 11x_2$ $s/t \ 2x_1 + x_2 \leq 104$ $x_1 + 2x_2 \leq 76$ $x_1, x_2 \geq 0$	$Max Z = 5x_1 + 7x_2$ $s/t \ x_1 + x_2 \leq 4$ $3x_1 + 8x_2 \leq 24$ $10x_1 + 7x_2 \leq 35$ $x_1, x_2 \geq 0$	$Min Z = 4x_1 + 2x_2$ $s/t \ x_1 + 2x_2 \geq 2$ $3x_1 + x_2 \geq 3$ $4x_1 + 3x_2 \leq 6$ $x_1, x_2 \geq 0$
33. A company manufactures two types of cloths, using three different colors of wool. One-yard length of type A cloth requires 4 Oz of red wool, 5 Oz of green wool and 3 Oz of yellow wool. One-yard length of type B cloth requires 5 Oz of red wool, 2 Oz of green wool and 8 Oz of yellow wool. The wool available for manufacturer is 1000 Oz of red wool, 1000 Oz of green wool and 1200 Oz of yellow wool. The manufacturer can make a profit of Rs.5 on one yard of type A cloth and Rs.3 on one yard of type B cloth. Find the best combination of the quantities of type A cloth and type B cloth which gives him maximum profit by solving the LPP graphically.		

34.	Solve the following LLP by Simplex method:		
	$Max Z = 7x_1 + 5x_2$ $s/t \quad x_1 + x_2 \leq 6$ $4x_1 + 3x_2 \leq 12$ $x_1, x_2 \geq 0$	$Max Z = 4x_1 + 10x_2$ $s/t \quad 2x_1 + x_2 \leq 50$ $2x_1 + 5x_2 \leq 100$ $x_1, x_2 \geq 0$	$Max Z = 6x_1 + 11x_2$ $s/t \quad 2x_1 + x_2 \leq 104$ $x_1 + 2x_2 \leq 76$ $x_1, x_2 \geq 0$
	$Max Z = 5x_1 + 3x_2$ $s/t \quad 3x_1 + 5x_2 \leq 15$ $5x_1 + 2x_2 \leq 10$ $x_1, x_2 \geq 0$	$Max Z = 3x_1 + 5x_2 + 4x_3$ $s/t \quad 2x_1 + 3x_2 \leq 18$ $2x_2 + 5x_3 \leq 18$ $3x_1 + 2x_2 + 4x_3 \leq 25$ $x_1, x_2, x_3 \geq 0$	$Max Z = -5x_1 + 8x_2 + 3x_3$ $s/t \quad 2x_1 + 5x_2 - x_3 \leq 1$ $-3x_1 - 8x_2 + 2x_3 \leq 4$ $-2x_1 - 12x_2 + 3x_3 \leq 9$ $x_1, x_2, x_3 \geq 0$
35.	Show that the following LPP does not have any feasible solution. Objective function for maximization $z = 20x + 30y$ subject to the constraints $3x + 4y \leq 24$, $7x + 9y \geq 63$ and $x \geq 0$, $y \geq 0$ by using simplex method.		
36.	Use simplex method to maximize $z = x + 1.5y$ subject to the constraints $x + 2y \leq 160$, $3x + 2y \leq 240$, given $x, y \geq 0$.		
37.	Minimize $P = x - 3y - 2z$ subject to the constraints $3x - y + 2z \leq 7$, $-2x + 4y \leq 12$, $-4x + 3y + 8z \leq 10$ given $x \geq 0$, $y \geq 0$, $z \geq 0$ by applying simplex method.		
38.	A factory uses 3 types of machines to produce two types of electronic gadgets. The first gadget requires i_1 hours 12, 4 and 2 respectively on the three types of machines. The second gadget requires i_2 hours 6, 10 and 3 on the machines respectively. The total available time in hours respectively on the machines 6000, 4000, 1800. If the two types of gadgets respectively fetches a profit of rupees 400 and 1000 find the number of gadgets of each type to be produced for getting the maximum profit by applying simplex method.		
39.	Solve the following LLP by Big M method:		
	$Min Z = 12x_1 + 20x_2$ $s/t \quad 6x_1 + 8x_2 \leq 100$ $7x_1 + 12x_2 \geq 120$ $x_1, x_2 \geq 0$	$Min Z = 2x_1 + 3x_2$ $s/t \quad x_1 + x_2 \geq 5$ $x_1 + 2x_2 \geq 6$ $x_1, x_2 \geq 0$	$Min Z = 4x_1 + 8x_2 + 3x_3$ $s/t \quad x_1 + x_2 \geq 2$ $2x_1 + x_3 \geq 5$ $x_1, x_2, x_3 \geq 0$
	$Min Z = 2x_1 + x_2$ $s/t \quad 3x_1 + x_2 = 3$ $4x_1 + 3x_2 \geq 6$ $x_1 + 2x_2 \leq 3$ $x_1, x_2 \geq 0$	$Max Z = 3x_1 + 2x_2$ $s/t \quad 2x_1 + x_2 \leq 2$ $3x_1 + 4x_2 \geq 12$ $x_1, x_2 \geq 0$	$Max Z = 3x_1 + 2x_2 + x_3$ $s/t \quad -3x_1 + 2x_2 + 2x_3 = 8$ $-3x_1 + 4x_2 + x_3 = 7$ $x_1, x_2, x_3 \geq 0$
40	Solve the following LPP using Big-M method, minimize $z = 2x_1 + x_2$, subject to the constraints $3x_1 + x_2 = 4$, $4x_1 + 3x_2 \geq 6$, $x_1 + 2x_2 \leq 3$, $x_1 \geq 0$, $x_2 \geq 0$.		

41.	Use Big-M method to minimize $z = 4x_1 + 3x_2$, subject to the constraints, $2x_1 + x_2 \geq 10$, $-3x_1 + 2x_2 \leq 6$, $x_1 + x_2 \geq 6$, $x_1 \geq 0$, $x_2 \geq 0$.		
42.	Solve the following LLP by Two phase method:		
	$Min Z = 2x_1 + 3x_2$ $s/t \quad x_1 + x_2 \geq 5$ $x_1 + 2x_2 \geq 6$ $x_1, x_2 \geq 0$	$Max Z = 3x_1 - x_2$ $s/t \quad 2x_1 + x_2 \geq 2$ $x_1 + 3x_2 \leq 3$ $x_2 \leq 4$ $x_1, x_2 \geq 0$	$Max Z = -2x_1 - x_2$ $s/t \quad 3x_1 + x_2 = 3$ $4x_1 + 3x_2 \geq 6$ $x_1 + 2x_2 \leq 4$ $x_1, x_2 \geq 0$
	$Min Z = 2x_1 + 3x_2$ $s/t \quad x_1 + x_2 \leq 5$ $x_1 + 2x_2 \geq 6$ $x_1, x_2 \geq 0$	$Max Z = 3x_1 + 2x_2$ $s/t \quad 2x_1 + x_2 \leq 2$ $3x_1 + 4x_2 \geq 12$ $x_1, x_2 \geq 0$	$Max Z = 5x_1 + 3x_2$ $s/t \quad 2x_1 + x_2 \leq 1$ $x_1 + 4x_2 \geq 6$ $x_1, x_2 \geq 0$
43.	Use two phase simplex method to minimize $Z = -2x_1 - x_2$ subject to the constraints $x_1 + x_2 \geq 2$, $x_1 + x_2 \leq 4$, $x_1, x_2 \geq 0$.		
44.	Use two phase simplex method to minimize $z = x_1 + x_2$, subject to the constraints $2x_1 + x_2 \geq 4$, $x_1 + 7x_2 \geq 7$, $x_1 \geq 0$, $x_2 \geq 0$.		